



Reverse alcohol desire

Scientists find brain circuit for alcohol cravings and could temporarily inactivate and reverse alcohol craving. <https://digit.in/apr19-9>



Acoustic metamaterial' cancels sound

A sound-blocking ringlike structure is created, for cutting out sounds while maintaining airflow. <https://digit.in/apr19-10>

FINDING THE CURE

Recently, scientists announced the second ever patient to be rid of HIV in history. Does this mean we have a cure?

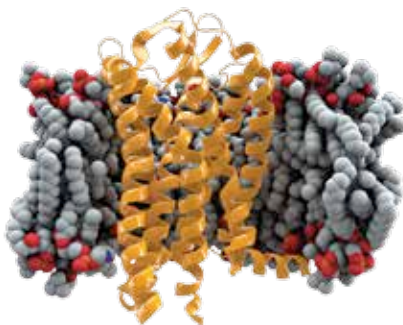
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Last month, a particular announcement shook the world of medical science - according to a group of researchers, a person had been cured of HIV. What a number of people also didn't realise was that this isn't the first time this was happening. Twelve years since the last patient was cured, scientists have seemingly repeated the feat and taken a huge leap towards developing an actual, effective cure for HIV that would benefit millions around the world. On the other hand, a segment of the scientific world has also cautioned people about the potential ill-effects and fallouts of this feat. To understand why this story has two sides at all, we need to look back at the Berlin patient.

THE BERLIN PATIENT

Timothy Brown made history when he revealed his identity as the Berlin Patient. Back in 1995, Brown was diagnosed with HIV and started antiretroviral therapy. What that means is

that he started taking medication to keep HIV from affecting him. In 2006, Timothy Brown was diagnosed with acute myeloid leukaemia. Dr Gero Hütter, the physician who was handling his case at the Charité Hospital in Berlin, got him signed up for a hematopoietic stem cell transplant. What



CCR5 receptor (yellow) in cell membrane (grey)

was interesting about this transplant was that it was from a donor who had the "delta32" mutation on the CCR5 receptor in their genes. The CCR5 protein gets mutated, and as a result, the majority of HIV cannot enter human

cells. Although Brown underwent a couple of transplants, he stopped his retroviral medication the day of the first transplant. Brown has since remained off HIV medication and is widely considered to be 'cured'. Which brings us to the London Patient.

THE LONDON PATIENT

12 years after Timothy Brown's treatment, at an AIDS conference in Seattle, scientists announced the London Patient, the only other patient to have been reportedly 'cured' of HIV following nearly the exact same steps as Timothy Brown's case. The London Patient has chosen not to disclose their identity, and have been off their medication since the last 18 months with no sign of HIV returning. Additionally, in both cases, cancer affecting the patients also went into remission. Neither of these people has to take the daily medications that people all around the world affected with HIV have to consume to live a regular, long life. So, why are we not celebrating the curing of HIV?